LINEAR MODELS · A SAMPLE BOOK FOR LEMMA

Least Squares in Twelve Pages

Projections, the normal equations and ridge regression

The Lemma authors

Read this with Lemma

This short book is here to try Lemma on. Every numbered statement, equation and figure can be previewed by resting the pointer on a reference such as Theorem 1.3 or (2.2), and peeked at by clicking it; Command-click a term such as *hat matrix* to look it up. Put the reading line in the proof of Proposition 3.3 and press F5 to step through it. In Section 3.1 the book says that a formula is "easily seen": select it and choose *Derive* to check each step yourself.

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Chapter 1

Projections

1.1 Orthogonality

Throughout, vectors are columns, $x^T y$ is the inner product of $x, y \in \mathbb{R}^N$ and $\|x\| = (x^T x)^{1/2}$ is the Euclidean norm. Two vectors are *orthogonal* when $x^T y = 0$; for orthogonal vectors the theorem of Pythagoras reads

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2. \tag{1.1}$$

Let X be an $N \times p$ matrix whose columns are linearly independent, so that $p \leq N$ and $X^T X$ is invertible. Its *column space* $\mathrm{col}(X)$ is the set of vectors $X\beta$ with $\beta \in \mathbb{R}^p$.

Definition 1.1. The *orthogonal projection* onto $\mathrm{col}(X)$ is the matrix

$$H = X(X^T X)^{-1} X^T. \tag{1.2}$$

In statistics H is called the *hat matrix*, because it puts the hat on y: $\hat{y} = Hy$.

Lemma 1.2. *The matrix H of* (1.2) *is symmetric and idempotent: $H^T = H$ and $H^2 = H$.*

Proof. The matrix $X^T X$ is symmetric, hence so is its inverse, and therefore $H^T = H$. Moreover

$$\begin{aligned} H^2 &= X(X^T X)^{-1} (X^T X) (X^T X)^{-1} X^T \\ &= X(X^T X)^{-1} X^T = H. \end{aligned}$$

□

Theorem 1.3 (Best approximation)**.** *For every $y \in \mathbb{R}^N$, the vector Hy is the point of* $\mathrm{col}(X)$ *closest to y:*

$$\|y - Hy\| \le \|y - v\| \quad \textit{for every } v \in \mathrm{col}(X),$$

with equality only for $v = Hy$.

Proof. Let $v = X\gamma$ be a point of $\mathrm{col}(X)$. The residual $y - Hy$ is orthogonal to every column of X, since $X^T(y - Hy) = X^T y - X^T X(X^T X)^{-1} X^T y = 0$ by (1.2). The vector $Hy - v$ lies in $\mathrm{col}(X)$, so it is orthogonal to $y - Hy$, and (1.1) gives

$$\|y - v\|^2 = \|y - Hy\|^2 + \|Hy - v\|^2. \tag{1.3}$$

The second term of (1.3) is positive unless $v = Hy$, which proves the claim. □

Figure 1.1 shows the case $p = 1$: projecting onto a line.

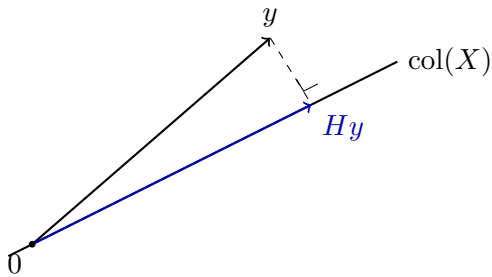


Figure 1.1: The orthogonal projection Hy of y onto a line, and the residual $y - Hy$, orthogonal to the line.

Chapter 2

Least squares

2.1 The normal equations

Given observations $y \in \mathbb{R}^N$ and the $N \times p$ matrix X of Chapter 1, the method of *least squares* (Legendre, 1805; Gauss, 1809) chooses $\beta \in \mathbb{R}^p$ to minimize

$$\mathrm{RSS}(\beta) = \|y - X\beta\|^2 = (y - X\beta)^T(y - X\beta). \tag{2.1}$$

Proposition 2.1. *A vector $\hat{\beta}$ minimizes* (2.1) *if and only if it solves the normal equations*

$$X^T X\hat{\beta} = X^T y. \tag{2.2}$$

Since $X^T X$ is invertible, the minimizer is unique:

$$\hat{\beta} = (X^T X)^{-1} X^T y. \tag{2.3}$$

Proof. By Theorem 1.3, $X\hat{\beta}$ minimizes $\|y - X\beta\|$ exactly when $X\hat{\beta} = Hy$. Multiplying $X\hat{\beta} = X(X^T X)^{-1}X^T y$ on the left by X^T gives (2.2); conversely, (2.2) gives (2.3), and then $X\hat{\beta} = Hy$ by (1.2). □

The same equations come from calculus: the gradient of (2.1) is

$$\frac{\partial\,\mathrm{RSS}}{\partial\beta} = -2X^T(y - X\beta), \tag{2.4}$$

and setting it to zero gives (2.2). The fitted values are $\hat{y} = X\hat{\beta} = Hy$, the projection of Definition 1.1.

2.2 Degrees of freedom

Definition 2.2. The *degrees of freedom* of a method whose fitted values are $\hat{y} = Sy$ for a fixed $N \times N$ matrix S is $\mathrm{df}(S) = \mathrm{trace}(S)$.

Theorem 2.3. *Least squares with p linearly independent columns has p degrees of freedom.*

Proof. By Definition 2.2 and the cyclic property of the trace,

$$\begin{aligned}\mathrm{trace}(H) &= \mathrm{trace}\left(X(X^TX)^{-1}X^T\right) = \mathrm{trace}\left((X^TX)^{-1}X^TX\right)\\ &= \mathrm{trace}(I_p) = p.\end{aligned}$$

□

2.3 Solving with a QR factorization

Forming X^TX squares the condition number of the problem, so in practice (2.2) is solved through a factorization of X itself (Golub and Van Loan, 2013; Trefethen and Bau, 1997).

Theorem 2.4. *Write $X = QR$, where Q is $N \times p$ with orthonormal columns ($Q^TQ = I_p$) and R is a $p \times p$ invertible upper triangular matrix (a QR factorization). Then*

$$\hat{\beta} = R^{-1}Q^Ty, \tag{2.5}$$

which is computed by back substitution in $R\hat{\beta} = Q^Ty$.

Proof. Since $Q^TQ = I_p$, we have $X^TX = R^TQ^TQR = R^TR$ and $X^Ty = R^TQ^Ty$. The normal equations (2.2) become $R^TR\hat{\beta} = R^TQ^Ty$, and multiplying by $(R^T)^{-1}$ gives $R\hat{\beta} = Q^Ty$. □

Example 2.5. Fitting a line $y \approx \beta_1 + \beta_2 t$ through the points $(0, 1)$, $(1, 2)$ and $(2, 4)$ uses

$$X = \begin{pmatrix}1 & 0\\ 1 & 1\\ 1 & 2\end{pmatrix}, \qquad y = \begin{pmatrix}1\\ 2\\ 4\end{pmatrix},$$

so that

$$X^TX = \begin{pmatrix}3 & 3\\ 3 & 5\end{pmatrix}, \qquad X^Ty = \begin{pmatrix}7\\ 10\end{pmatrix},$$

and (2.3) gives $\hat{\beta} = (5/6,\ 3/2)^T$: the line $y = 5/6 + 3t/2$.

Chapter 3

Regularization

3.1 Ridge regression

When the columns of X are nearly dependent, X^TX is nearly singular and (2.3) is unstable. *Ridge regression* (Hoerl and Kennard, 1970) adds a penalty with a parameter $\lambda > 0$:

$$\mathrm{PRSS}(\beta) = \|y - X\beta\|^2 + \lambda\|\beta\|^2. \tag{3.1}$$

Proposition 3.1. *The minimizer of* (3.1) *is easily seen to be*

$$\hat{\beta}_\lambda = (X^TX + \lambda I)^{-1}X^Ty. \tag{3.2}$$

Proof. As in (2.4), the gradient of (3.1) is $-2X^T(y - X\beta) + 2\lambda\beta$. It vanishes when $(X^TX + \lambda I)\beta = X^Ty$, and $X^TX + \lambda I$ is invertible because $\lambda > 0$; the criterion is strictly convex, so this point is its minimum. □

The penalty *shrinks* the coefficients towards zero. The singular value decomposition shows by how much, direction by direction.

3.2 The singular value decomposition

Theorem 3.2 (Eckart and Young, 1936)**.** *The matrix X can be written*

$$X = UDV^T, \tag{3.3}$$

where U is $N \times p$ with orthonormal columns $u_1, \ldots, u_p$, V is a $p \times p$ orthogonal matrix and $D = \mathrm{diag}(d_1, \ldots, d_p)$ with $d_1 \geq \cdots \geq d_p > 0$, the singular values of X.

Proposition 3.3. *The fitted values of ridge regression are*

$$X\hat{\beta}_\lambda = \sum_{j=1}^{p} u_j \frac{d_j^2}{d_j^2 + \lambda} u_j^T y. \tag{3.4}$$

Proof. Substitute (3.3) into (3.2). Since $V^T V = I$ and $U^T U = I$,

$$X^T X + \lambda I = V D^2 V^T + \lambda V V^T = V(D^2 + \lambda I)V^T. \tag{3.5}$$

Hence, by (3.5) and (3.2), $\hat{\beta}_\lambda = V(D^2 + \lambda I)^{-1} V^T V D U^T y = V(D^2 + \lambda I)^{-1} D U^T y$. Then

$$X\hat{\beta}_\lambda = U D V^T V (D^2 + \lambda I)^{-1} D U^T y = U D (D^2 + \lambda I)^{-1} D\, U^T y,$$

and since the matrices in the middle are diagonal this is (3.4). □

Corollary 3.4. *Ridge regression has*

$$\mathrm{df}(\lambda) = \sum_{j=1}^{p} \frac{d_j^2}{d_j^2 + \lambda} \tag{3.6}$$

degrees of freedom. It decreases as λ grows and tends to p as $\lambda \to 0$, in agreement with Theorem 2.3.

Proof. By (3.4) the fitted values are Sy, where S is the matrix $UD(D^2 + \lambda I)^{-1} DU^T$. By Definition 2.2 and the cyclic property of the trace,

$$\mathrm{trace}(S) = \mathrm{trace}\left(D(D^2 + \lambda I)^{-1} D\right) = \sum_{j=1}^{p} \frac{d_j^2}{d_j^2 + \lambda},$$

which is (3.6). Each term decreases in λ and tends to 1 as $\lambda \to 0$. □

Exercises

Exercise 3.1. Show that (3.2) is the least-squares solution (2.3) for the augmented data $\tilde{X} = \begin{pmatrix} X \\ \sqrt{\lambda} I \end{pmatrix}$ and $\tilde{y} = \begin{pmatrix} y \\ 0 \end{pmatrix}$.

Exercise 3.2. Using (3.3), show that $\|\hat{\beta}_\lambda\|$ decreases as λ grows.

Exercise 3.3. Verify Example 2.5 with Theorem 2.4, by computing a QR factorization of its X.

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